

Evertrust ERP — System Overview

AI-assisted tender operations: what the platform does, and how n8n and Claude fit around it.

Snapshot: 2026-05-31 · repo: evertrust-ERP · branch: m0-foundations

Status: **LIVE** = built and deployed · **PARTIAL** · **PLANNED** = in the docs, not built yet. Where the canonical docs describe more than exists, the gap is called out.

1 · What this system is

Two halves, joined at the moment a **deal is won**:

- **REACH ARSENAL** (pre-deal acquisition) — outbound sales running autonomously in **n8n Cloud on OpenAI** (gpt-4o / gpt-4o-mini) + Gmail / Sheets / Drive / Calendar / WhatsApp. Targets companies in a niche/city, cold-emails them, sorts replies, books meetings. *Not Claude, not the ERP.*
- **Evertrust ERP** (post-deal operations) — **the web-app**. Once a deal is won, the tender becomes an ERP record and runs the German procurement (Vergabe) pipeline: intake -> pricing -> approval -> documents -> submit -> result.

Stack: NestJS (Drizzle + Postgres 16, JWT/argon2) · Next.js (App Router, shadcn/ui, TanStack Query) ·

@evertrust/shared (Zod single source of truth) · multi-tenant · permission-RBAC · immutable audit ·

Claude-only for AI · n8n Cloud. Roadmap = the canonical 52-step / 8-phase tender workflow (R01–R52).

2 · ERP web-app functions

Identity and access — LIVE

- JWT + argon2 login; roles **SUPER_ADMIN** > **ADMIN** > **MANAGER** > **EMPLOYEE** + a 21-permission RBAC catalog, enforced server-side and used to gate the UI.
- **Per-user permission overrides** — the API re-reads role/permissions from the DB on every request, so edits and deactivations take effect immediately.
- **/users page**: role / position (CEO ... Officer) / department editing, deactivate, per-user permissions.
- **Immutable audit log** on every mutation (actor + before/after, org-scoped).

Tender core — LIVE (Phase 4)

- Tender CRUD keyed by the portal **Vergabe-ID**; the **7-status state machine** (NOT_STARTED -> PIC_PRICING -> CUSTOMER_PRICING -> DOCUMENTS -> SUBMITTED -> AWARDED -> LOST), guarded and audited.
- Manual **PIC assignment**; **TYPE 1 document** upload / list / download; supplier and customer registries.

Pricing workbench — LIVE (Phase 5, highest value)

- **5a — LV line items + PriceObservation engine**: evidence from 7 weighted sources; per-line REAL / MIXED / ESTIMATE signal -> **R/Y/G**; confidence scoring; tender **high-risk** rule (>=35% unbacked, or any top-5-by-value line unbacked). Finalize -> CUSTOMER_PRICING.
- **5b — Claude price-assist**: 'Ask Claude' returns a unit-price **suggestion** (confidence + rationale + assumptions). Never auto-applies — a human accepts it as an AI_ESTIMATE (line stays unbacked/RED until a real quote backs it). Logged to ai_runs; low confidence escalates.
- **5c — Hermes supplier RFQ**: 'Request quotes' fires the Hermes n8n webhook to email suppliers; dispatch logged; replies return as SUPPLIER_QUOTE evidence.

Approval and deadline — LIVE (Phase 6)

- **6a — Customer-approval gate**: 'no written approval -> no submission', enforced in code; evidence is channel-agnostic (link or note).

- **6b — Deadline safety:** deterministic risk (T-2 / T-1 / T-0 escalate to MANAGER / ADMIN / SUPER_ADMIN; reminders T-5 / T-3 / T-1) -> dashboard card, per-tender badge, /tenders/deadline-risk worklist.

QC and submission — LIVE (Phase 7 core, R34–R37)

- **Conditional QC gate:** required before submission when above the EU threshold OR pricing is high-risk OR a QC review was opened; routine tenders skip it (reuses the QC approval type).
- **Submit act + evidence:** POST /tenders/:id/submit enforces every blocker, snapshots the file list, writes an immutable **submission receipt** (proof + timestamp + files), and advances DOCUMENTS -> SUBMITTED. A direct transition to SUBMITTED is refused, so **SUBMITTED <=> a logged receipt**. The portal act stays human.

Growth Engine — LIVE (the ERP's Arsenal control panel)

- /growth-engine launches **AIM** campaigns (fires the AIM n8n webhook -> provisions the Drive campaign).
- **'Run now'** per Arsenal stage + a dependency-free **daily Bazooka scheduler** (send time + timezone editable in-app). Every ERP->n8n hand-off logged in arsenal_runs (DISPATCHED / FAILED).

Not built in the app

- **PARTIAL** — TYPE 2 doc checklist / completeness (R32–R33): needs the external EPC Document Library manifest. (TYPE 2 files already upload via the Phase 4 documents module.)
- **PLANNED** — Phase 2 intake, Phase 3 shortlist, Phase 8 result/follow-up.

3 · The 8-phase tender workflow (R01–R52)

Phase	What it covers	Status
1 — Partner scouting	Out of automation scope (Kha's lane)	FROZEN
2 — Search + intake	Argus portal search (TED / DTVP / Service-Bund) + Scribe GAEB X81/X83 + OCR	PLANNED
3 — Shortlist	Sieve match vs client profiles -> confirm/reject	PLANNED
4 — Record + assign	Tender record, PIC assignment, TYPE 1 docs	LIVE
5 — Pricing	LV + evidence engine + Claude assist + Hermes RFQ	LIVE
6 — Approval + deadline	Customer-approval gate + deadline risk	LIVE
7 — Docs + QC + submit	QC gate + submit + evidence (R34-R37); TYPE 2 checklist (R32-R33)	PARTIAL
8 — Result + follow-up	Win/loss, contract, billing, supplier review	PARKED

Domain anchors: regimes VOB/A, VgV, UVgO · GAEB DA XML (X81 unpriced LV, X83 priced response, X84 final) · portal submission stays human · 'submit at T-2' · GDPR/EU.

4 · How n8n fits

- **Live today:** the REACH ARSENAL project — 7 workflows (AIM, Lead Satellite, Ammo Forge, Reach Bazooka, Reply Glock, Sleeper Grenade, Niche Flamethrower) on **OpenAI**. The OpenAI key is an n8n credential, not an ERP setting.
- **ERP -> n8n (works):** the ERP fires webhooks for AIM launch, per-stage 'Run now', the daily Bazooka send, and the Hermes RFQ. Each is config-gated — a blank webhook URL = that trigger is safely off.
- **n8n -> ERP (writeback): absent** — the ERP is localhost-only, so n8n cannot reach it yet. The one intended pull is read-only GET /tenders/deadline-risk ('n8n polls'); nothing polls it yet.
- **Planned tender-ops workflows** (Argus / Scribe / Sieve) that would feed the ERP are Phase 2/3 — not built.

5 · How Claude fits

- **Strategy (COMBINE):** Claude-only for ERP AI (DeepSeek dropped; CLAUDE/DEEPSEEK survive only as audit-enum labels). BUILD_PLAN positions Claude as a supervisor / QA layer (~10-30%), activating on low confidence, red flags, or sensitive items.
- **Live today:** exactly one integration — price-assist (5b) — calling the Anthropic Messages API directly (raw fetch, forced structured tool output, no SDK). Disabled until ANTHROPIC_API_KEY is set.
- **Mismatch to keep in view:** the docs say 'Claude', but the live automations (the Arsenal) run on OpenAI. Separate worlds — the ERP does not call OpenAI, and the Arsenal does not call Claude.

Agent	Intended role	Status
Argus	Tender portal search / intake	PLANNED
Scribe	GAEB / PDF parsing -> structured fields	PLANNED
Sieve	Bid/skip matching vs client profiles	PLANNED
Hermes	Supplier / customer outreach (RFQ)	PARTIAL
Hydra	Orchestration	PLANNED
Eve	Review / QA (price-assist is the closest live piece)	PLANNED
Nero / Aza / Cipher	Code / docs / security	PLANNED

6 · Bottom line

Production-ready and deployed — the middle of the tender pipeline: record -> price (with Claude assist + supplier RFQ) -> customer approval -> deadline safety -> QC -> submit with evidence, plus full auth / RBAC / user management and the Growth-Engine control panel. All multi-tenant, permission-gated, audited.

Gaps vs the docs' full vision — automated intake/parsing/shortlisting (Argus/Scribe/Sieve, Phases 2-3), result/follow-up (Phase 8), the broader Claude agent fleet, and n8n->ERP writeback (blocked until the ERP is deployed to a reachable URL). Today a human starts each ERP phase; the automations assist and escalate rather than run end-to-end.

Operating principle: automation assists, humans decide. Low confidence and missing approvals escalate rather than proceed; every critical action is audited; nothing reaches SUBMITTED without recorded customer approval, conditional QC, and a submission receipt.